

DOCUMENT 05

Backend and persistence integration

Additive architecture, versioned masks and unchanged legacy behavior

1. Treat this as a policy branch, not a global refactor

Confirmed: preserve the old option. **Proposed:** introduce an explicit `CleaningAreaMode` with legacy and `text_shape` variants. Missing fields in old records resolve to legacy.

Do not change existing global growth constants, morphology behavior or engine defaults to make the new mode work. Route `text_shape` through a dedicated mask-plan builder and leave the old geometry path intact. The current core already has pixel masks, patches and a bounded compositor that can be extended. [S07, S11, S12]

Independent settings

Axis	Proposed values	Compatibility rule
Cleaning area	legacy / text_shape	Old records and unchanged defaults use legacy.
Detection profile	existing / enhanced_sfx	Optional detector availability does not change mask semantics.
Text policy	legacy_japanese / all_detected_text	Only the latter bypasses recognition/script gating.
Outside bubbles	review / clean	Explicit scope decision; not inferred from an engine choice.
Engine	Existing supported choices	New mode adds no automatic cloud or heavy-engine escalation.

A convenience preset may choose `all_detected_text` plus outside-bubble cleaning after an explicit user action. It must show those choices and preserve a reproducible run snapshot.

Proposed extension points

New module: `text_mask` or equivalent, owning refinement, footprint derivation and immutable `MaskPlan` creation. The name is proposed; it does not currently identify an inspected file.

Existing modules: `detect/balloon` for candidates; `fit` for background statistics and routing; engines for bounded generation; `composite` for final enforcement; `project/patch` for artifacts and provenance. See the audited source map in Document 02.

Do not represent automatically generated text masks as hand-drawn solely to skip growth. Their source, behavior and provenance are different even when both have authoritative boundaries.

2. Data model and storage proposal

Keep object identity separate from geometry

Changing padding must not create a different region ID. Store an identity plus a revision. Bbox is the current storage/indexing extent, not the region's permanent identity or erase membership.

```
TextRegionPlan {
  region_id, source_id, source_revision,
  cleaning_area_mode, algorithm_version,
  detector_provenance[],
  candidate_geometry, enclosure, fragment_group_id?,
  base_mask_ref, base_mask_hash,
  applied_mask_ref, applied_mask_hash,
  alpha_ref?, alpha_hash?,
  padding_source_px, grouping_options, feather_source_px,
  refinement_context, engine_context,
  mask_quality_state, correction_revision,
  plan_revision
}
```

These are proposed fields, not the app's current schema. Integrate with existing sidecar buffers and provenance rather than duplicating the project store.

Keep three artifacts distinguishable

Base mask: lettering before the user's growth settings. It supports idempotent padding and correction.

Applied support: the exact allowed edit footprint that the preview displays. It is the compositor's mask.

Generated patch: engine pixels blended only within support, stored in the source's supported color/depth representation.

The current ApiRegion/ApiMask layer describes region geometry and provenance; it needs an additional reference to the actual prepared footprint for text-shaped previews. [\[S14\]](#)

Serialization and migration

Version the settings and mask-plan payload. Missing fields use legacy behavior; unknown future modes fail visibly rather than falling back to a rectangle. Existing patches can continue to render from stored buffers without loading any newly proposed model.

Converting an old region is explicit: recompute from the preserved source, preview the new plan and commit only after acceptance. Never relabel an old padded/applied mask as the new immutable base.

Persist enough to reproduce the chosen plan even if a model's defaults later change. Store detector and segmenter checkpoint hashes independently from the inpainting model hash.

3. Preview/apply protocol and state transitions

Proposed commands

prepare_text_mask(region, options, expected_source_revision) returns a prepared plan, mask preview reference, resolved parameters, quality state and plan hash.

apply_prepared_mask(plan_id, expected_plan_hash, engine) verifies source and plan identity, runs the engine with bounded geometry and commits a patch through W.

update_text_mask(plan_id, corrections, options) builds a new revision from the stored base or explicit edited base, not from the last applied support.

Command names are illustrative. Reuse the app's established command/event style and transport security rather than adding a parallel framework.

State vocabulary

```
candidate -> preparing -> ready_for_preview -> applying -> cleaned
               -> needs_mask_correction
candidate -> held_by_policy
applying  -> declined_by_engine / failed / cancelled
```

A preview artifact exists before a cleaned patch. Do not use “has patch” as the only way to represent a valid candidate mask. Keep skipped detections reviewable.

Race and failure behavior

Padding or source changes invalidate in-flight plans. Late model responses must not overwrite a newer revision. Cancellation before commit leaves the current visible layer unchanged. If preparing an optional specialist fails, explain the fallback detector profile without changing cleaning-area mode.

An engine cannot silently widen W to repair a quality failure. It can retry with another supported engine using the same plan, or offer a new, visibly larger plan for explicit acceptance.

Preview transport

Prefer a local mask artifact or tiled alpha texture addressed by region and revision rather than full-page binary arrays in JSON. Return a page-coordinate bbox plus mask dimensions/origin. Keep short-lived file/protocol access within existing local security boundaries.

The mask texture is authoritative; simplified contours are a display cache. The renderer must not resample a contour and then use that approximation to perform backend writes.

4. Engines, re-runs and compatibility

Engine integration checklist

Fill: sample sufficient clean surrounding paper, but paint only W. Sampling and write area are independent.

Denoise: it may inspect surrounding pixels, but any generated smoothing outside W must be discarded. Do not leave the current margin arithmetic implicitly active in text_shape.

LaMa: feed a suitable context crop and the planned hole. The existing narrower ink and isolation logic must be mode-aware; it cannot add an unpreviewed ring. [S08, S09]

Other enabled per-region engines: enforce the same final write support, honor existing permission and availability rules, or decline text_shape explicitly. This feature does not introduce an automatic cloud upload.

Quality gates: assess residual text and reconstruction without widening the mask as a hidden side effect. No OCR success/failure becomes an eligibility condition.

Re-running a layer

Remove the layer being replaced from the context composite. Regenerate its geometry from the stored base and corrections. Replacing a larger old mask with a smaller new one must reveal original/lower-layer pixels outside the new footprint.

Engine changes alone do not grow the mask. Undo restores both the patch and exact plan/settings revision. Redo restores the same artifacts, or a clearly recorded rerender when exact artifacts are unavailable.

Export and long strips

The displayed and exported page must use the same applied support. Lossless PNG/TIFF pixel comparisons can test outside-mask equality; do not use recompressed JPEG byte equality as the test.

Respect source/decode-window boundaries. Carry transforms through strip joins and split outputs back to the correct pages. A virtual gap or unverified join must never become an editable pixel region. Preserve current color mode, bit depth, profile and metadata handling; unsupported engine combinations remain explicit declines.

Rollback

Keep legacy path tests and feature flags. Disabling enhanced discovery must not disable text_shape. Disabling text_shape must not rewrite completed patches. Saved results remain viewable without the optional models that created them.

Source IDs link directly to primary references. Full evidence ledger: Document 08.